🧬 ProteinForge: The Complete Idea

Course: UE23EC343BC1 – Deep Learning Capstone
Team: 4 Members | **Timeline:** 8 Weeks
Dataset: ProteinGym β-lactamase
Philosophy: Build incrementally. Each tier is complete. Stop whenever time runs out.

🎯 The Core Problem (What We're Solving)

The Real-World Challenge

Drug companies want to know:

> "If I mutate this protein, will it work better or worse?"

Why this matters:

- Testing ONE mutation in the lab costs $500-1000
- Takes 2-4 weeks
- Proteins have thousands of positions × 20 amino acids = millions of possibilities
- **Can't test them all experimentally**

Our solution: Use AI to predict which mutations are good BEFORE going to the lab.

💡 The Intuitive Idea (ELI5)

Imagine You're Breeding Dogs

Traditional approach:

```
1. Breed two dogs → Wait 6 months → See if puppies are good
2. Repeat thousands of times
3. Very slow, very expensive
```

Our AI approach:

```
1. Look at 1000s of dog photos and their traits (data)
2. Learn: "Floppy ears + Short legs = Good temperament"
3. PREDICT what puppies will look like BEFORE breeding
```

```
4. Only breed the ones we predict will be good
5. Much faster, much cheaper
```

Replace "dogs" with "proteins" and that's our project.

🧬 ProteinForge: The 3-Tier Rocket

We build the project in **3 independent stages**. Each stage is a complete, working system.

🥉 TIER 1: The Predictor (Weeks 1-3)

What it does: Given a protein mutation, predict if it's good or bad

The Problem:

```
Input:  "Protein X, position 42, change A→G"
Output: "Fitness score = 0.85" (good!) or "0.23" (bad!)
```

How It Works (Multi-Modal Learning):

Think of it like judging a book:

Method 1: Read the text (Sequence)

- Look at the amino acid sequence: "MKTIIALSYIFCL..."
- Like reading the words in a book
- **Model:** ESM-2 (pre-trained protein language model)
- Gives us: "Does this sequence make sense?"

Method 2: Look at the structure (Structure)

- How does the protein fold in 3D?
- Like looking at the book's layout and formatting
- **Model:** 2D CNN on contact maps (distance between amino acids)
- Gives us: "Is this structure stable?"

Method 3: Check the evolution (Evolution)

- What positions are conserved across species?
- Like checking if a word appears in ancient texts (must be important!)
- **Model:** MLP on conservation scores
- Gives us: "Is this position critical?"

Fusion Network:

```
Sequence features (1280 numbers)
Structure features (256 numbers)    → Combine → Final Score
Evolution features (128 numbers)
```

What We Achieve:

- **Metric:** Spearman correlation (ρ) between our predictions and real lab data
- **Target:** ρ ≥ 0.65 (competitive with published methods)
- **Baseline to beat:** Simple ESM-2 alone gets ρ ~ 0.50

Why This Guarantees Grade A:

- ✅ Covers all 4 syllabus units
- ✅ Demonstrates multi-modal learning
- ✅ Reproducible results
- ✅ Clear evaluation metrics
- ✅ Even if we STOP here, it's a complete capstone project

🥈 TIER 2: The Generator (Weeks 4-5)

What it does: INVENT new mutations that we predict will be good

The Upgrade:

```
Tier 1: "You give me a mutation, I tell you if it's good"
Tier 2: "I SUGGEST mutations that I think are good"
```

How It Works (Generative Modeling):

The Idea:

- Train a VAE (Variational Autoencoder) on successful mutations
- The VAE learns a "map of good mutations"
- We can explore this map to find new, untested mutations

Think of it like:

```
Tier 1 = Google Maps (you give address, it shows rating)
Tier 2 = Restaurant recommendation (it SUGGESTS good places)
```

The Loop:

```
1. Generator creates 100 new mutation ideas
2. Run them through Tier 1 Predictor (our judge)
3. Keep the top 20 that score highest
4. Use those to train Generator to make even better ones
5. Repeat
```

This is called: Guided generation (like GANs, but for proteins)

What We Achieve:

- Generate 100 novel mutations
- **Target:** 20% of them score higher than wild-type (original protein)
- Demonstrate diversity (not just generating the same mutation 100 times)

Why This Makes It Better:

- ✅ Now we're not just analyzing, we're DESIGNING
- ✅ Shows understanding of generative AI (VAE)
- ✅ Demonstrates closed-loop optimization
- ✅ "Generative AI for drug design" looks amazing on resume
- ✅ Still a complete project if we STOP here

🥇 TIER 3: The Physicist (Weeks 6-8)

What it does: Validate our AI predictions using REAL PHYSICS

The Upgrade:

```
Tier 1+2: "AI thinks these mutations are good"
Tier 3: "Physics CONFIRMS they actually are good"
```

The Problem We Solve:

Question: How do we know the AI isn't bullshitting?

Answer: Test its predictions against computational physics!

How It Works (Physics Validation):

We take the top 20 mutations from Tier 2 and run them through:

Option A: AlphaFold Confidence (Easiest)

```
1. Fold the mutated protein with ESMFold API
2. Get confidence score (pLDDT)
3. Question: Do high-fitness predictions = high-confidence structures?
```

Option B: FoldX Stability (Medium)

```
1. Use FoldX to compute ΔΔG (stability change)
2. Question: Do high-fitness predictions = stable proteins?
```

Option C: AutoDock Binding (Hardest)

```
1. Dock substrate (antibiotic) to mutated protein
2. Measure binding energy
3. Question: Do high-fitness predictions = strong binding?
```

We pick based on time available.

The Key Result:

```
Correlation between:
- AI fitness predictions (from Tier 1)
- Physics validation scores (from Tier 3)

Target: ρ ≥ 0.45 (shows AI learned something real about physics)
```

What We Achieve:

- **Primary:** Show AI predictions correlate with physics ($\rho \geq 0.45$)
- **Bonus:** Find 3-5 mutations that are validated as stable/functional
- **Story:** "Our AI doesn't just predict—it understands real chemistry"

Why This Makes It Publication-Worthy:

- ✅ Most ML papers stop at prediction
- ✅ We go further: prediction → generation → physics validation
- ✅ Demonstrates END-TO-END understanding
- ✅ Shows system can replace expensive lab work
- ✅ Physics validation = concrete proof of utility

🎯 Why This Design Works

1. **Incremental Risk**

```
Tier 1 risk: 10% ← Safe, known techniques
Tier 2 risk: 30% ← VAEs can be tricky
Tier 3 risk: 60% ← Physics tools can break

Total project risk: LOW (because tiers are independent)
```

The genius: If Tier 3 fails, we still have Tier 1+2 (Grade A+)

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2. Multiple Exit Points

```
Stop at Tier 1: Grade A, solid capstone
Stop at Tier 2: Grade A+, impressive capstone
Finish Tier 3: Grade A+, publication potential
```

You can't fail. Each tier is a complete achievement.

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3. Natural Progression

```
Tier 1: Discriminative AI (Classification/Regression)
   ↓
Tier 2: Generative AI (VAE)
   ↓
Tier 3: Physics-Informed AI (Validation)
```

This tells a story: From basic ML → Advanced ML → Real-world validation

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4. Syllabus Coverage Gets Stronger

```
Tier 1: Covers all 4 units (minimally)
Tier 2: Covers all 4 units (solidly)
Tier 3: Covers all 4 units (impressively)
```

Even Tier 1 alone passes the course requirements.

🧪 The Technical Stack (What We Actually Use)

Data:

- **Source:** ProteinGym β-lactamase dataset
- **Size:** ~5,000 protein variants with experimental fitness scores
- **Format:** CSV (sequence, mutation, fitness score)

Models:

Tier 1 Models:

```
Sequence Branch:  ESM-2 (650M params, pre-trained)
Structure Branch: 2D CNN on contact maps
Evolution Branch: MLP on conservation scores
Fusion:           Attention + MLP
```

Tier 2 Model:

```
Generator: VAE (LSTM encoder/decoder)
Training:  KL-divergence + reconstruction loss
Guidance:  Tier 1 model as fitness oracle
```

Tier 3 Tools:

```
Folding:   ESMFold API (free)
Stability: FoldX (free, CPU)
Docking:   AutoDock Vina (optional)
```

Tech:

- Python 3.9+
- PyTorch (deep learning)
- BioPython (protein handling)
- Pandas/NumPy (data processing)

📊 Success Metrics (How We Know It Works)

Tier 1 Metrics:

```
✅ Spearman ρ ≥ 0.65 (prediction accuracy)
✅ Beat ESM-2 baseline (ρ ~ 0.50)
✅ Ablation studies show each modality helps
```

Tier 2 Metrics:

```
✅ Generate 100 unique, valid sequences
✅ 20+ score higher than wild-type (in Tier 1 model)
✅ Show diversity (not mode collapse)
```

Tier 3 Metrics:

```
✅ AI predictions ↔ Physics scores: ρ ≥ 0.45
✅ 3-5 mutations validated as stable/functional
✅ Better than random mutations
```

🎓 What This Teaches

Technical Skills:

- Multi-modal deep learning
- Transfer learning (ESM-2)
- Generative modeling (VAE)
- Optimization algorithms
- Physics-informed ML
- Systems integration

Soft Skills:

- Incremental development
- Risk management
- Project scoping
- Research thinking
- Scientific storytelling

Career Skills:

- "I built an AI-physics pipeline for drug design"
- Shows versatility: discriminative + generative + validation
- Demonstrates ability to complete complex projects
- Publication-quality work (if Tier 3 completes)

🏆 The Final Picture

What Recruiters See:

```
"This student:
 ✓ Built a multi-modal protein fitness predictor
 ✓ Extended it to generate novel designs
 ✓ Validated predictions with computational physics
 ✓ Showed systems thinking (3 components working together)
 ✓ Demonstrated research maturity (incremental, safe design)

→ Hire this person."
```

What Professors See:

```
"This team:
 ✓ Covered all 4 syllabus units thoroughly
 ✓ Went beyond requirements (generative + physics)
 ✓ Followed scientific methodology
 ✓ Produced reproducible results
 ✓ Potential for publication

→ Grade: A+"
```

What You Learn:

```
"I now understand:
 ✓ How to build production ML systems
 ✓ How to manage project risk
 ✓ How AI connects to real-world validation
 ✓ How to do research incrementally
 ✓ How to tell a compelling technical story

→ I'm ready for industry or grad school."
```

🎬 The Bottom Line

ProteinForge is:

- ✅ A safe way to do ambitious work
- ✅ Three projects in one (each tier standalone)
- ✅ Guaranteed grade + optional publication
- ✅ Maximum learning with minimum risk

It works because:

- Each tier builds on the previous (but doesn't depend on it)

- Risk increases only when foundation is solid
- Multiple exit points = can't fail
- Tells a complete story: predict → generate → validate

This is how real research works:

1. Start with solid foundation
2. Add innovation carefully
3. Validate before claiming success
4. Stop when you run out of time

You're not building a project. You're building a system with built-in safety nets.

That's ProteinForge. Now go build it. 🚀